

Existential "Amount" Relatives (4/16/00 - comments welcome)¹

1. Introduction

Existential relatives (ERs) like (1) are relative clauses that have gaps in the postcopular position of *there*-existential constructions. ERs have been assumed by Carlson (1977), Heim (1987), and Grosu and Landman (1998) to be members of a class of relative clauses distinct from restrictive and appositive relatives - the class of amount relatives (ARs), which includes examples like (2) under the reading that denotes the *quantity* (e.g., the number of liters) of champagne that they spilled that evening:

- (1) the books [(that) there were ___ on the table]
- (2) the champagne [(that) they spilled ___ that evening] (quantity reading)

ERs have been assumed to be ARs for two major reasons. FIRST, both (1) and (2) are incompatible with overt WH-relativizers:

- (3) *the books [*which* there were ___ on the table]
- (4) *the champagne [*which* they spilled ___ that evening] (quantity reading)

SECOND, both (1) and (2) are compatible with only a limited class of determiners on their heads (see §3.3 for discussion):

- (5) *the/*most/*some* books [(that) there were ___ on the table]
- (6) *the/*most/*some* champagne [(that) they spilled ___ that evening] (quantity reading)

Despite the similarities in (3-6), I argue that ERs are not ARs, for two major reasons:

- (7) a. The quantity readings that distinguish ARs from restrictive relatives are not possible in ERs; i.e., (1) cannot denote the *quantity* of books on the table².
- b. ERs and ARs are not compatible with the same classes of determiners on their heads. In particular, ERs are compatible with (non-distributive) universals, but members of the relevant subclass of ARs are not.

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² Cf. Grosu and Landman (1998: 132-134), McNally (1992: 139-140).

I propose, following suggestions by McNally (1992, 1998), that the relativizer and determiner restrictions on ERs are derived from the semantics of *there*-existential constructions, not the semantics of amounts and degrees.

This paper is structured as follows. In §2, I compare the differences between RRs and ERs to the differences between RRs and ARs. In §3, I critically review previous analyses of ERs and argue against them, based on the observations in (7). In §4, I propose a property-argument analysis of ERs. In particular, following McNally's (1998) claims that *FIRST*, the argument of a *there*-existential construction is of type $\langle e, t \rangle$ and that *SECOND*, the addition of an existential sentence to a context entails the introduction of a unique discourse referent into the domain of the context that corresponds to the instantiation of the property that this argument denotes, I propose that existential relatives have gaps that are of type $\langle e, t \rangle$ and do not combine semantically with their head nominals through set intersection or predicate modification, but through function application. Furthermore, I propose that the discourse referent in ERs is introduced into the domain of the *content* by Existential Disclosure (cf. Dekker (1993), Chierchia (1995)). In §5, I discuss the consequences of these proposals for the analysis of RRs that involve *whose* and have the same determiner restriction as ERs, but no relativizer restriction and conclude that an adequate theory of the relation between relativization and *there*-existential constructions should separate the theoretical principle responsible for uniqueness/maximality from the theoretical principle responsible for the type of the relativizer and the gap in ERs.

2. Background

2.1. Restrictive Relatives vs. Existential Relatives

It is standardly assumed that restrictive relatives (RRs) are intersective modifiers of nominals and denote sets of individuals which semantically combine with their head nominal through set intersection ((8)b) or predicate modification ((8)c)³:

- (8) a. (the) books [that __ were on the table]
 b. $\{x \mid x \text{ is a book}\} \cap \{x \mid x \text{ was on the table}\}$
 c. $\lambda x(x \text{ is a book} \wedge x \text{ was on the table})$

The representations in (8)b and (8)c denote the set of things such that they are both books and on the table (ignoring tense). On the surface, this also seems like a reasonable way to represent the semantics of ERs (set intersection in (9)a; predicate modification in (9)b):

- (9) a. (the) books [(that) there were __ on the table]
 b. $\{x \mid x \text{ is a book}\} \cap \{x \mid \text{there was } x \text{ on the table}\}$
 c. $\lambda x(x \text{ is a book} \wedge \text{there was } x \text{ on the table})$

The RR in (8)a and the ER in (9)a seem to be equivalent in that a sentence containing (8)a and a sentence containing (9)a are *ceteris paribus* logically equivalent.

³ My version of predicate modification is as follows:

(i) If α is a branching node, $\{\beta, \gamma\}$ the set of α 's daughters, $\llbracket \beta \rrbracket = \lambda x[P(x)]$, and $\llbracket \gamma \rrbracket = \lambda x[Q(x)]$, then $\llbracket \alpha \rrbracket = \lambda x[P(x) \wedge Q(x)]$.

But there are a number of grammatical restrictions that apply to ERs but not to RRs. FIRST, RRs are compatible with overt WH-relativizers, but ERs are not (cf. Carlson (1977: 526)):

- (10) a. the books [*which* __ were on the table]
 b. *the books [*which* there were __ on the table] [= (3)]
- (11) a. the people [*who* __ were at the party]
 b. *the people [*who* there were __ at the party]

SECOND, RRs are compatible with any determiners on their heads, but ERs are compatible with only a limited class (cf. Carlson (1977: 521)):

- (12) a. *the/most/some* books [that were __ on the table]
 b. *the/*most/*some* books [(that) there were __ on the table] [= (5)]

THIRD, RRs are compatible with negation of their predicates, but ERs are not. As far as I know, this has not been observed before:

- (13) a. the books [that __ weren't on the table]
 b. *the books [(that) there weren't __ on the table]

Thus, ERs are different from RRs in at least three distinct ways. But given these differences, why are the representations in (9) so similar to the representations in (8)? It seems that an adequate theory of ERs should predict these differences as consequences of the way that ERs are represented. But how should ERs be represented? The way that previous analyses have answered this question has been to assume that ERs are members of a class of relative clauses that seems to differ from RRs in the same ways - the class of ARs.

2.2. Restrictive Relatives vs. Amount Relatives

Examples like the relative clause in (14) are assumed (cf. Heim (1987: 38), Grosu and Landman (1998: 132)) to be prototypical ARs:

- (14) It will take us forever to drink
 [_{DP} the champagne [_{CP} (that) they spilled __ that evening]].

Note that the DP in (14) permits two readings:

- (15) a. *Identity of Substance (I of S)*
 The DP denotes the entity constituted by the champagne they spilled that evening.
 b. *Identity of Quantity (I of Q)*
 The DP denotes the quantity of champagne they spilled that evening.

Example (14) is ambiguous because the CP can be either an RR or an AR:

- (16) a. *RR (I of S)*
 $\lambda x[x \text{ is champagne} \wedge \text{they spilled } x \text{ that evening}]$
 'the set of entities that are constituted by champagne and were spilled by them that evening'
- b. *AR (I of Q)*
 $\lambda d[\text{they spilled } d\text{-much champagne that evening}]$
 'the set of degrees such that they spilled that much champagne that evening'

Note that this analysis of the ambiguity suggests a certain amount of polymorphism in the semantics of the definite determiner *the*. In particular, it suggests that *the* is ambiguous between denoting a function from sets of individuals to individuals and denoting a function from sets of degrees to degrees. I assume that this is *sortal* polymorphism; i.e., ambiguity between functions of the same type. The definite determiner that applies to (16)a and the definite determiner that applies to (16)b are both functions of type $\langle\langle e, t \rangle, e\rangle$, where D_e is the domain of all entities (individuals, degrees, etc.). In (16)a, *the* is of sort $\langle\langle e, t \rangle, e\rangle$, where D_e is the domain of individuals and $D_e \subseteq D_e$. In (16)b, *the* is of sort $\langle\langle d, t \rangle, d\rangle$, where D_d is the domain of degrees and $D_d \subseteq D_e$. In both cases, *the* is a function from an ordered domain to its maximal element - in Boolean terms, its supremum⁴.

There are a number of grammatical restrictions that apply to ARs but not to RR. FIRST, RRs are compatible with overt WH-relativizers, but ARs are not (cf. Carlson (1977: 526)):

- (17) the champagne [*which* they spilled __ that evening]
 (I of S/*I of Q)

SECOND, RRs are compatible with any determiners on their heads, but ARs are compatible with only a limited class (cf. Carlson (1977: 521)):

- (18) a. *the* champagne [(that) they spilled __ that evening]
 (I of S/I of Q)
- b. *most/some* champagne [(that) they spilled __ that evening]
 (I of S/*I of Q)

THIRD, RRs are compatible with negation of their predicates, but ARs are not⁵. As far as I know, this has not been observed before:

- (19) the champagne [(that) they didn't spill __ that evening]
 (I of S/*I of Q)

⁴ To not complicate the discussion further than necessary, I use 'type' instead of 'sort' below.

⁵ It is hard to separate the I of S readings from the I of Q readings in examples like (19). But note that in a context like 14, examples like (i) that have only I of Q readings are not compatible with negation, as in (ii):

- (i) It will take us forever to earn the money [(that) Bill Gates spent __ on his security system last year]
 (ii) *It will take us forever to earn the money [(that) Bill Gates didn't spend __ on his security system last year]

Thus, ARs are different from RRs in at least three distinct ways. Note that these are very similar to the ways in which ERs are different from RRs. Given these similarities, should the representations of ERs be similar to the representations of ARs - i.e., should they involve abstraction over degrees instead of individuals? Or is there a third alternative?

2.3. The Definiteness Restriction

If relative clauses involve variable binding, it seems reasonable to assume that they have variables in their gaps. Heim observes that there are restrictions on the types of variables in the gaps of ERs. In particular, she observes that (9)b and (9)c violate the definiteness restriction in (20):

- (20) *Definiteness Restriction* (Heim (1987: 23))
 *There be x , where x is an individual variable.

This restriction accounts for several special properties of English *there*-existential constructions.

FIRST, definite nominals and pronominal variables are ungrammatical in this context:

- (21) a. There is a book on the table.
 b. *There is *The Brothers Karamazov* on the table.
 c. *There is it on the table.

Overt names like *The Brothers Karamazov* in (21)b are classified with individual variables under the assumption that the semantics of names is essentially analogous to that of individual variables, except that the latter depend on a variable assignment and the former do not.

SECOND, wide-scope readings for nominal phrases in the pivot of existential *there* are not possible. If a nominal phrase that has wider scope than another (non-c-commanded) element in a sentence is moved and has its place filled by a variable at LF, then the wide scope reading of (22)b is ruled out by (20). Representations for the relevant readings are as in (23):

- (22) a. Ralph believed that an FBI agent was spying on him.
 (wide/narrow scope of *an FBI agent*)
 b. Ralph believed that there was an FBI agent spying on him.
 (*wide/narrow scope of *an FBI agent*)
- (23) a. *Wide Scope*
 $\exists x[\text{FBIagent}(x) \wedge \text{believe}(\text{Ralph}, \text{'spying.on}(x, \text{Ralph})\text{'})]$
 'There was an FBI agent x such that Ralph believed x was spying on him.'
 b. *Narrow Scope*
 $\text{believe}(\text{Ralph}, \text{'}\exists x[\text{FBIagent}(x) \wedge \text{spying.on}(x, \text{Ralph})\text{'})]$
 'Ralph believed there was an FBI agent x such that x was spying on him.'

In the wide scope reading, the existential quantifier binds an individual variable x in the scope of *believe*. In the narrow scope reading, the existential quantifier *is* in the scope of *believe*. Thus in (22)b, the wide scope reading is not possible because it involves an individual variable in the pivot of existential *there*.

THIRD, wide-scope readings for nominal phrases extracted from the pivot of existential *there* are not possible. If an extracted nominal phrase has its place filled by a variable at LF, then the wide scope reading of (24)b is also ruled out by (20). Representations for the relevant readings are as in (25); $\llbracket d\text{-many FBI agents} \rrbracket = \lambda x[\text{FBI.agent}(x) \wedge |x| = d]$:

- (24) a. How many FBI agents did Ralph believe were spying on him?
(wide/narrow scope of *how many FBI agents*)
b. How many FBI agents did Ralph believe there were spying on him?
(*wide/narrow scope of *how many FBI agents*)
- (25) a. *Wide Scope*
 $\lambda p \exists d \llbracket p = \text{'}\exists x[\text{FBI.agent}(x) \wedge |x| = d \wedge \text{believe}(\text{Ralph}, \text{'spying.on}(x, \text{Ralph}))]\text{'}$
'For what d were there d -many FBI agents x such that Ralph believed x were spying on him?'
b. *Narrow Scope*
 $\lambda p \exists d \llbracket p = \text{'believe}(\text{Ralph}, \text{'}\exists x[\text{FBI.agent}(x) \wedge |x| = d \wedge \text{spying.on}(x, \text{Ralph}))]\text{'}$
'For what d did Ralph believe there were d -many FBI agents spying on him?'

Assuming that there is a way of decomposing a phrase like *how many FBI agents* into a [*for what d*] part and a [*d -many FBI agents*] part (cf. Cresti (1995: 83-88)), the representations in (25) suggest the following explanation. In the wide scope reading, *d-many FBI agents* binds an individual variable x in the the scope of *believe*. In the narrow scope reading, *d-many FBI agents* (which is not interpreted as an individual variable) *is* in the scope of *believe*. Thus in (24)b, the wide scope reading is not possible because it involves an individual variable in the pivot of existential *there*.

Why is this relevant? *How many*-questions are some of the best-understood cases of WH-extraction involving non-individual variable binding and thus may suggest an explanation for the grammaticality of non-individual variable binding in ERs and ARs. Carlson (1977), Heim (1987), and Grosu and Landman (1998) all assume that the relevant non-individual variables in ERs are variables over degrees (i.e., variables of type d). A variable of type d is compatible with *that* and $\emptyset$, but not with overt WH-relativizers⁶. If both ERs and ARs are compatible with *that* and $\emptyset$, but not with overt WH-relativizers, then an explanation involving degree variable binding in both ERs and ARs seems reasonable. If this is on the right track, the relativizer restriction has the following explanation (adapted from Grosu and Landman (1998: 128-129)):

- (1) the books [(that) there were on the table]

⁶ This may be why (i) and (ii) are ungrammatical:

- (i) *the rides [*how many* Max went on at Disney World]
(ii) *the money [*how much* Max spent at Disney World]

- (3) *the books [*which* there were ___ on the table]
- (20) *Definiteness Restriction*
*There be *x*, where *x* is an individual variable.
- (26) *Unacceptability of (3)*
- The gap of relativization with relativizer *which* is filled by an individual variable.
 - This means that (3) contains an individual variable in a position which is subject to (20), hence (3) is ungrammatical.
- (27) *Acceptability of (1) - first try*
- There is a way to avoid the problem in (3), namely *degree abstraction*: the complementizers *that* and $\emptyset$ (but not the relativizer *which*) are compatible with a degree variable.
 - The gap in (1) contains a null degree expression, *d-many books*, in which only the variable *d* is bound by the relativizer.
 - The degree variable is not in a position subject to (20), and the null degree expression *d-many books*, which is, does not count as an individual variable.
 - Thus there is no definiteness problem in (1); (1) is grammatical.

But there is also another way to explain the relevant facts. Individual variables (i.e., variables of type *e*) are compatible with *that*, $\emptyset$, and overt WH-relativizers. If neither ERs nor ARs are compatible with overt WH-relativizers, then an explanation involving non-individual variable binding in both ERs and ARs seems reasonable. Note that this similarity between ERs and ARs does not mean that they involve variables of the same type. In §4.2, I suggest that they have similar relativizer restrictions as a consequence of involving two different sorts of variables - property variables (type $\langle e, t \rangle$) in the case of existential relatives, and degree variables (type *d*) in the case of amount relatives.

3. Previous Analyses

The three previous analyses that I discuss here do not analyze ERs and ARs as separate classes, but instead assume that the possibility of relativization of the position after *there be* is really the best diagnostic for ARs⁷. I critically review these analyses and observe that they are

⁷ There are a number of different subclasses of ARs that I do not discuss, in particular antecedent-contained deletion (ACD) examples like (i):

- (i) a. Max ate all [(that) he could ___].
b. Max ate everything [(that) he could ___].

Note that (i)a has a reading where Max ate to satiation, and that (i)b has both a reading where Max ate to satiation and a reading where Max ate all of the allowable foods but may have remained quite hungry. Furthermore, note that the former reading (the quantity reading) is excluded with negation and with root modals:

- (ii) a. Max ate everything [(that) he couldn't ___].
b. Max ate everything [(that) he should ___].

consistent with the claim that the gaps in ERs are not of type *e*, but ultimately either predict that identity of quantity readings are possible in ERs (Carlson (1977), Heim (1987)) or that ERs and ARs are compatible with the same classes of determiners on their heads (Grosu and Landman (1998)).

3.1. Carlson (1977) and Heim (1987)

Carlson's (1977: 522-524) general approach is to derive the relativizer and determiner restrictions on ERs from a comparative-like syntax. He suggests that the analysis of ERs such as (28)b parallels the analysis of comparatives (equatives) such as (28)a. In particular, he assumes the underlying forms in (29)a-b for (28)a-b:

- (28) a. as many books [as there were ___ on the table]
 b. the books [(that) there were ___ on the table] [= (1)]
- (29) a. **[NP [QP [DET as [S' as there were [NP [QP [DET THAT][Q AMOUNT]]][_{NOM} books]] on the table]]**[Q AMOUNT]][_{NOM} books]]
 b. **[NP [QP [DET the [S' (that) there were [NP [QP [DET THAT][Q AMOUNT]]][_{NOM} books]] on the table]]]**[_{NOM} Ø]]

In (29)a, the boldfaced NP is deleted and the S' is extraposed to the end of the main NP to derive the surface order in (28)a. In (29)b, the boldfaced QP is deleted, [_{NOM} books] is raised into head position (i.e., [_{NOM} Ø]), and the S' is extraposed to the end of the main NP to derive the surface order in (28)b. The relativizer restriction follows from Carlson's approach because, like the analysis of comparatives in (29)a, "[t]he analysis proposed [in (29)b] does not allow for the creation of forms such as *who* or *which* in the derivation, since the underlying relativized NP is not of the same structure as corresponding NP's in RR's" (Carlson (1977: 526)). Note that both (29)a and (29)b suggest compositionality puzzles. In (29)a, it is unclear why [_{NOM} books] is in the boldfaced QP. In (29)b, it is unclear why [_{NOM} books] raises from the pivot of existential *there* into head position.

It is also unclear why the determiner restriction follows from Carlson's approach. He says, "[l]et us for the moment use relativization of the logical subject of an S that has undergone *There*-Insertion as our chief diagnostic for ascertaining what range of determiners and quantifiers may co-occur with AR's, and what range may not". Ignoring that this begs the question with respect to whether ERs are ARs, consider (30) and (31):

Example (i)a, which has only the quantity reading, is ungrammatical with negation and with root modals:

- (iii) a. *Max ate all [(that) he couldn't ____].
 b. *Max ate all [(that) he should ____].

I have no explanation for the compatibility of quantity readings with universal determiners in these examples.

- (30) *Class I Determiners* (may cooccur with ERs; from Carlson (1977: 525(10)))
- The* people [(that) there were ___ at that time] only lived a few decades.
 - That's *all* [(that) there is ___].
 - What/That* light [(that) there is ___ in this painting] is quite diffuse.
 - Any*⁸ beer [(that) there may be ___ left in the cooler] is mine.
 - Every* lion [(that) there is ___] eats meat.
- (31) *Class II Determiners* (may not cooccur with ERs; from Carlson (1977: 525(11)))
- *Five/*Most/*Several/*Many* men [(that) there were ___ here] disagreed.
 - *Some/*Each/*A* man [(that) there was ___] disagreed.

Class I includes definites (*the*, non-deictic *that*), some universals (*every*, *all*) free choice *any*, and *what*. Class II includes indefinites (*some*, *a*), cardinal determiners (*five*, *several*, *many*), *most*, and *each*. Carlson says:

I think that the analysis offered here can lead us to an understanding of why these particular items appear in Class I, rather than in some other arbitrary grouping. Recall that, in the analysis of the comparatives, it is assumed that a binding relation of some sort exists between the [*as*] determiner of the matrix and the *THAT* determiner of the subordinate clause. Both of these items may precede some expression of amount in underlying structure (the Q AMOUNT in [(29)a]). If we assume that the relationship between [*as*] and *THAT* may hold only between elements of the same syntactic category, and that the same sort of relationship is found in amount relatives between the quantifier or determiner of the matrix and the *THAT* in the relative clause, then it should follow that the Class I elements must be of the syntactic category of those elements that precede an expression of amount, like [*as*]. (Carlson (1977: 525); my italics)

Thus, Carlson analyzes the difference between Class I and Class II determiners as a syntactic (i.e., subcategorical) difference. Exactly how this analysis relates to the semantic differences between Class I and Class II determiners is unclear.

Heim (1987) follows Carlson's suggestion that the analysis of ERs parallels the analysis of comparatives, but her general approach is to derive the relativizer and determiner restrictions on ERs from a comparative-like *semantics*. Comparative clauses like the bracketed constituent in (28)a denote degree descriptions; comparative morphemes like *-er* and *as* denote relations between degrees. Thus, (28)a denotes the degree d' such that d' is equal to the maximal degree in $\lambda d[\text{there were } d\text{-many books on the table}]$. Heim suggests that the bracketed constituents in both (28)a and (28)b are represented (approximately) as in (32), where MAX is defined as in (33):

- (32) $\text{MAX}(\lambda d[\text{there were } d\text{-many books on the table}])$

⁸ Free choice *any* seems to behave differently in ERs than in RRs. For example, (i) seems to mean that you may eat some of the apples in the box, but (ii) seems to mean that you may eat *all* of the apples in the box (thanks to Bill Ladusaw for observing the relevance of these facts):

- You may eat *any* apples [that ___ are in the box].
- You may eat *any* apples [(that) there are ___ in the box].

- (33) *Maximality* (Rullmann (1995: 55))
 For D a set of degrees, $\leq$ a total order on D ,
 $\text{MAX}(D) = \lambda d \in D. [\forall d' \in D. [d' \leq d]]$.

There is a problem with Carlson and Heim's analyses of ERs. The representation in (32) denotes the maximal degree in a set of degrees - in the plural case, the maximal number in a set of numbers. But Grosu and Landman (1998: 132-134) and McNally (1992: 139-140) observe that from a number, the actual sum of individuals that number is a number of cannot be reconstructed. Thus, the Carlson/Heim analysis predicts that ERs can only support identity of quantity readings. But ERs can only support identity of substance readings. For example, (34)a can only be paraphrased as (34)b:

- (34) a. the people [(that) there were ___ at the party]
 b. 'the entity constituted by the people there were at the party'
 c. *'the cardinality of the entity constituted by the people there were at the party'

The reading in (34)c is not possible, thus the Carlson/Heim analysis cannot be correct. I discuss two ways to solve this problem (the *quantity/substance problem*). On the one hand, the degree analysis of ERs can be accepted, but a richer notion of degree can be developed. On the other hand, the degree analysis of ERs can be rejected. Grosu and Landman (1998) take the former approach, discussed in §3.2; McNally (1992) takes the latter approach, discussed in §4.1.

3.2. Grosu and Landman (1998)

Grosu and Landman (1998) assume that the Carlson/Heim analysis is correct in most aspects; for example, they accept the explanation for the relativizer restriction in (26-27). They argue that the quantity/substance problem can be solved if "a notion of degree that keeps track of what it is a degree of" (Grosu and Landman (1998: 134)) is introduced. A "classical degree" d , as in (35)a, is the cardinality of a plural individual. A "structured degree" ∂ relative to a sortal predicate P , as in (35)b, is an ordered triple of the cardinality of a plural individual, the sortal predicate, and the plural individual itself:

- (35) a. For all plural individuals x : $d = |x|$.
 b. For all plural individuals x : $\partial = \langle |x|, P, x \rangle$.

Grosu and Landman propose that the notion of degree in (35)b is on the right track⁹. Maximality is defined as in (36):

- (36) For P a set of plural individuals, D a set of structured degrees, $\leq$ a partial order on P ,
 a. $\text{SUP}(P) =$
 $\lambda x \in P. [\forall x' \in P. [x' \leq x]]$.
 b. $\text{max}(D) =$
 $\langle \lambda \text{SUP}(\lambda x \in P. [\langle |x|, P, x \rangle \in D]) \lambda, P, \text{SUP}(\lambda x \in P. [\langle |x|, P, x \rangle \in D]) \rangle$.

⁹ To not complicate the discussion further than necessary, I translate Grosu and Landman's set-theoretic representations into function-theoretic ones. SUP is their supremum operator.

In the case of ERs, the relevant sum of individuals is reconstructed as follows. FIRST, MAX applies to $\max(D)$, giving the singleton set containing the maximal degree in $\max(D)$. MAX is defined as in (37):

- (37) For D a set of structured degrees, $\partial \in D$,
 $\text{MAX}(\partial) =$
 $\lambda \partial [\partial = \max(D)]$ if $\max(D) \in D$, undefined otherwise.

SECOND, SUBSTANCE applies to $\text{MAX}(\max(D))$, giving the singleton set containing the plural individual in the ordered triple in $\text{MAX}(\max(D))$. SUBSTANCE is defined as in (38):

- (38) For P a set of plural individuals, D a set of structured degrees,
 $\text{SUBSTANCE}(D) =$
 $\lambda x \in P. [\langle \lambda x, P, x \rangle \in D]$.

Thus, the bracketed constituent in (39)a is represented as in (39)b:

- (39) a. the books [(that) there were ___ on the table] [= (1)]
 b. $\text{SUBSTANCE}(\text{MAX}(\max(\lambda \partial [\text{there were } \partial\text{-many books on the table}])))$

The representation in (39)b denotes the singleton set that contains the plural individual in the ordered triple that is the maximal degree in $\lambda \partial [\text{there were } \partial\text{-many books on the table}]$. Thus, the actual sum of individuals that a number is a number of can be reconstructed. But there is a problem with this part of the analysis. If there are no constraints on the application of SUBSTANCE (Grosu and Landman (1998: 141) claim that it applies "in the unmarked case"), then the prediction is that universals (and other determiners that "preserve maximality") support identity of quantity readings because SUBSTANCE is a function from sets of degree triples to sets of plural individuals that is not sensitive to the relevant sortal distinctions (e.g., individuals vs. degrees). But universals do not support identity of quantity readings:

- (40) a. every soldier (that) the Chinese paraded last May Day
 b. 'the entity constituted by the soldiers that the Chinese paraded last May Day'
 c. *'the cardinality of the entity constituted by the soldiers that the Chinese paraded last May Day'

The reading in (40)c is not possible, thus this part of the analysis cannot be correct.

Ignoring the problem above, how does this analysis explain the determiner restriction? Grosu and Landman (1998: 143-147) observe that only universals, definites, and partitives built from definites are grammatical in ERs. They claim that the determiner restriction follows from the definitions in (36-38). The relevant constraint is that ERs can only combine with determiners that preserve the information in (39)b - in particular, maximality - into the generalized quantifier meaning. Class I determiners (e.g., *all*, *the*) preserve maximality; Class II determiners (e.g., *most*, *three*) do not. The representations in (41) and the definitions in (42) make the notion of "preserving maximality" more explicit:

- (41) a. *All/The* books on the table were blue.
 $\llbracket \text{books on the table} \rrbracket \cap \llbracket \text{blue} \rrbracket = \llbracket \text{books on the table} \rrbracket$
 b. *Most/Three* books on the table were blue.
 $\llbracket \text{books on the table} \rrbracket \cap \llbracket \text{blue} \rrbracket \subseteq \llbracket \text{books on the table} \rrbracket$
- (42) A determiner $\mathcal{D}$ preserves maximality iff for all tripartite quantificational structures $\mathcal{D}(A)(B)$, $A \cap B = A$.

Thus, the determiners in (41)a preserve maximality because the intersections of the denotations of their restrictions and the denotations of their nuclear scopes are the same as the denotations of their restrictions. Furthermore, the determiners in (41)b do not preserve maximality because the intersections of the denotations of their restrictions and the denotations of their nuclear scopes are subsets of the denotations of their restrictions. As (40) suggests, the determiners that preserve maximality are roughly the determiners in Class I; i.e., definites and (most) universals. The determiners that do not preserve maximality are roughly the determiners in Class II. But there is also a problem with this part of the analysis; it ignores distributivity. This is relevant because the distributive universal determiners *each* and *every* are members of different classes - *every* is Class I, but *each* is Class II. Both *each* and *every* preserve maximality according to (42). But ERs are not compatible with *each*:

- (43) a. **each* book [(that) there was ___ on the table]
 b. *every* book [(that) there was ___ on the table]

Example (43)a is ungrammatical, thus the determiners that preserve maximality are not exactly the Class I determiners.

The upshot of this discussion is that there are two major problems with Grosu and Landman's analysis. FIRST, their proposals about how the actual sum of individuals that a number is a number of can be reconstructed incorrectly predict that ARs are compatible with all Class I determiners. SECOND, the determiners that preserve maximality are not exactly the Class I determiners. In §3.3, I discuss a new analysis of the determiner restriction that accounts for the facts in (40) and (43).

3.3. The Determiner Restriction

ACCOUNTING FOR (43) [more clear]

Both *each* and *every* preserve maximality according to (42). But *every* has a "collective universal construal" (cf. Beghelli and Stowell (1997: 98)) and groups with *all* but not *each* in ERs. *Every* also groups with *all* but not *each* in examples like (44):

- (44) a. It took **each/every* boy to lift the piano.
 b. One boy ate almost **each/every* apple.
 c. Not **each/every* boy ate an apple.

What is a "collective universal construal" in formal terms? Consider example (45):

- (45) Chuck, Friz, and Bob carried the piano upstairs.

On the collective reading of (45), Chuck, Friz, and Bob are agents of the same carrying event (i.e., the piano was carried upstairs once); on the distributive reading, Chuck, Friz, and Bob are agents of different carrying events (i.e., the piano was carried upstairs three times). Both *each* and *every* have distributive readings, but only *every* has a collective reading. Why? Nominals headed by *each* and *every*, like definites, other universals, and (conjunctions of) proper names, denote principal filters (cf. Barwise and Cooper (1981: 209)):

- (46) A generalized quantifier $\mathcal{Q}$ is a *principal filter* iff there is a set of individuals A such that A is not necessarily empty and for any set of individuals X , $X \in \mathcal{Q}$ iff $A \subseteq X$.

A is the *generator set* of $\mathcal{Q}$. For example, if Chuck, Friz, and Bob are the relevant boys, then the set $\{\text{Chuck}, \text{Friz}, \text{Bob}\}$ is the generator set of both $\llbracket \text{each boy} \rrbracket$ and $\llbracket \text{every boy} \rrbracket$. In the spirit of Partee (1987: 129-131), I assume that the collective reading of a nominal that denotes a principal filter $\mathcal{Q}$ is derived by a mapping LINK such that for $\{a, b, \dots, z\} = A$, A the generator set of $\mathcal{Q}$, $\text{LINK}(A) = \{a \oplus b \oplus \dots \oplus z\}$ ¹⁰. In formal terms, *every boy* can have a "collective universal construal", and *each boy* cannot, because the singleton set $\{\text{Chuck} \oplus \text{Friz} \oplus \text{Bob}\}$ can be the generator set of $\llbracket \text{every boy} \rrbracket$ (i.e., $\text{LINK}(\{\text{Chuck}, \text{Friz}, \text{Bob}\})$ is allowed), but not of $\llbracket \text{each boy} \rrbracket$ (i.e., $\text{LINK}(\{\text{Chuck}, \text{Friz}, \text{Bob}\})$ is not allowed). Thus, it seems reasonable to say that Class I determiners are all and only those determiners $\mathcal{D}$ such that $\mathcal{D}(A)$ is a principal filter that can have a plural singleton generator set, and that Class II determiners are all and only those determiners $\mathcal{D}$ such that $\mathcal{D}(A)$ is not a principal filter that can have a plural singleton generator set. Thus, "preserving maximality" is redefined as follows:

- (47) A determiner $\mathcal{D}$ *preserves maximality* iff for a set of individuals A , $\mathcal{D}(A)$ is a principal filter that can have a plural singleton generator set.

The same difference that exists between *each* and (non-distributive) *every* exists between *both* and *the two*:

- (48) a. *the two books* [(that) there were ___ on the table]
b. **both books* [(that) there were ___ on the table]

Both $\llbracket \text{both books} \rrbracket$ and $\llbracket \text{the two books} \rrbracket$ denote principal filters, but only $\llbracket \text{the two books} \rrbracket$ can have a plural singleton generator set (cf. Ladusaw (1982: 236-239)), thus only (48)a is grammatical.

ACCOUNTING FOR (40) [less clear]

The discussion above relates to the discussion of the polymorphism of *the* in §2.2. In particular, it suggests an explanation for why ERs and ARs are not compatible with the same classes of determiners on their heads. Note that for a set A , $\text{LINK}(A)$ is allowed only if the sum operation ' $\oplus$ ' is defined for the elements of A . This is the case only if the extension of A can be a join semilattice (cf. Link (1983: 306)). The extension of the domain D_e of individuals can be a join semilattice; i.e., there can be atomic individuals. But the extension of the domain D_d of

¹⁰ Under my definition, LINK denotes a function of type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$. Under Partee's definition, LINK denotes a function of type $\langle\langle e, t \rangle, e \rangle$ such that for $\{a, b, \dots, z\} = A$, A the generator set of $\mathcal{Q}$, $\text{LINK}(A) = a \oplus b \oplus \dots \oplus z$.

degrees cannot be a join semilattice; i.e., there cannot be atomic degrees. In both the individual and the degree cases, *the* is a function from an ordered domain to its maximal element - in Boolean terms, its supremum. But the sum operation ' $\oplus$ ' is defined for a set *A* only if the extension of *A* can be a join semilattice, and only the supremum of a join semilattice is a join. If only *the* is polymorphic in the relevant way; i.e., if only *the* can denote functions from sets of non-plural entities to non-plural entities, then only *the* is compatible with ARs.

3.4. Summary

The observations below, introduced in §1, are problematic for previous analyses of ERs. Carlson (1977) and Heim (1987) fail to account for (7)a, and Grosu and Landman (1998) fail to account for (7)b:

- (7) a. The quantity readings that distinguish ARs from restrictive relatives are not possible in ERs; i.e., (1) cannot denote the *quantity* of books on the table.
 b. ERs and ARs are not compatible with the same classes of determiners on their heads. In particular, ERs are compatible with (non-distributive) universals, but members of the relevant subclass of ARs are not.

The relevant facts about RRs, ERs, and ARs are summarized in (46):

(49)	RRs	ERs	ARs
Overt WH-words	✓	*	*
Definite heads	✓	✓	✓
Universal heads	✓	✓	*
Negated predicates	✓	*	*
Quantity readings	*	*	✓

In §4, I discuss a new analysis that accounts for the observations in (7). I basically ignore ARs and discuss the differences between ERs and RRs: the relativizer, determiner, and negation restrictions, and reconstruction effects.

4. A Property-Argument Analysis

I propose, following suggestions by McNally (1992, 1998), that the relativizer and determiner restrictions on existential relatives are derived from the semantics of *there*-existential constructions, not the semantics of amounts and degrees. In particular, following McNally's (1998) claims that FIRST, the argument of a *there*-existential construction is of type $\langle e, t \rangle$ and that SECOND, the addition of an existential sentence to a context entails the introduction of a unique discourse referent into the domain of the context that corresponds to the instantiation of the property that the argument denotes, I propose that existential relatives have gaps that are of type $\langle e, t \rangle$ and do not combine semantically with their head nominals through set intersection or predicate modification, but through function application. Furthermore, I propose that the discourse referent in ERs is introduced into the domain of the *content* by Existential Disclosure (cf. Dekker (1993), Chierchia (1995)). The consequences of these proposals for the analysis of ERs are summarized in (50):

- (50) a. The relativizer restriction follows because variables that are not of type e are not compatible with overt WH-relativizers.
 b. The determiner restriction follows because the unique discourse referent gives a plural singleton generator set for the generalized quantifier meaning of a nominal phrase containing an ER.
 c. The negation restriction follows because negation is opaque for binding of this discourse referent.
 d. Reconstruction effects follow from the proposal that ERs are of type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ and combine semantically with their head nominals through function application.

4.1. McNally (1992, 1998)

Informally stated, McNally's (1992) proposal is the following:

The existential predicate in English is interpreted as a property of a *description of an entity*, specifically the property that the description is instantiated by some entity at some index. The addition of a (non-negative, non-modalized) existential sentence to a context entails the introduction of a discourse referent into the domain of the context that corresponds to the instantiation of the description-argument. An additional felicity condition requires this referent to be novel. (McNally (1992: 4))

I follow McNally (1998) instead and interpret the existential predicate in English as a predicate of a property that entails that the property is instantiated by some entity in some situation. Under this assumption, the addition of a (non-negative, non-modalized) existential sentence to a context entails the introduction of a discourse referent into the domain of the context that corresponds to the instantiation of the property argument. Thus, (51)a is interpreted as in (51)b:

- (51) a. There are books on the table.
 b. $\text{there.be}(\lambda x[\text{book}(x) \wedge \text{on.the.table}(x)])$

There be is interpreted as synonymous with the predicate *to be instantiated (in some situation)*, which holds of expressions interpreted as properties; i.e., expressions of type $\langle e, t \rangle$. A nominal phrase is licit in the postcopular position of the English *there*-existential iff it has a well-defined property-type denotation or denotes a quantifier over properties. Following Partee (1987), if all nominal phrases (NPs, DPs, or whatever) have type $\langle\langle e, t \rangle, t\rangle$ denotations, then a nominal phrase α has a well-defined type $\langle e, t \rangle$ denotation iff the type-shifting function BE applied to α does not yield the empty set as its output. BE is defined as in (52):

- (52) $\text{BE}(\alpha) = \lambda x[\alpha(\lambda y[y = x])]$, where α is a generalized quantifier.

What is the upshot of these assumptions? In §4.2, I show that a McNally-like analysis suggests a different explanation for the relativizer restriction on ERs. In §4.3 and §4.4, I motivate the representation for ERs suggested in §4.2 with facts about the determiner and negation restrictions and reconstruction effects.

4.2. The Relativizer Restriction

If relative clauses in general denote functions involving abstraction over variables that are of type τ , as in (50), then ERs denote functions involving abstraction over variables that are of type $\langle e, t \rangle$, as in (51):

$$(53) \quad \llbracket [\text{CP } \text{Op}_i [\text{IP } \dots t_i \dots]] \rrbracket = \lambda \alpha \in D_e [\dots \alpha \dots] \quad (\text{type } \langle \tau, t \rangle)$$

$$(54) \quad \llbracket [\text{CP } \text{Op}_i (\text{that}) [\text{IP } \text{there were } t_i \text{ on the table}]] \rrbracket = \lambda P \in D_{\langle e, t \rangle} [\text{there.be}(\lambda x \in D_e . [P(x) \wedge \text{on.the.table}(x)])] \quad (\text{type } \langle \langle e, t \rangle, t \rangle)$$

Under this assumption, the relativizer restriction has the following explanation:

- (1) the books [(that) there were __ on the table]
- (3) *the books [*which* there were __ on the table]
- (20) *Definiteness Restriction*
*There be x , where x is an individual variable.
- (55) *Acceptability of (1) - second try*
 - There is a way to avoid the problem in (3), namely *property abstraction*: the complementizers *that* and $\emptyset$ (but not the relativizer *which*) are compatible with a property variable.
 - The gap in (1) contains a property variable, P , which is bound by the relativizer.
 - The property variable is in a position subject to (20), but (trivially) does not count as an individual variable.
 - Thus there is no definiteness problem in (1); (1) is grammatical.

But there is a problem with (54). If $\llbracket [\text{CP } \text{Op}_i (\text{that}) \text{there were } t_i \text{ on the table}] \rrbracket$, which is of type $\langle \langle e, t \rangle, t \rangle$, combines with $\llbracket [\text{NP books}] \rrbracket$, which is of type $\langle e, t \rangle$, through function application, then $\llbracket [\text{NP books } \text{Op}_i (\text{that}) \text{there were } t_i \text{ on the table}] \rrbracket$ is of type t . We don't want that; we want a function of type $\langle e, t \rangle$ that can combine with $\llbracket [\text{DP the}] \rrbracket$, which is of type $\langle \langle e, t \rangle, e \rangle$. The representation in (56) is consistent with what we want:

$$(56) \quad \lambda P \in D_{\langle e, t \rangle} . [\lambda y \in D_e . [\text{there.be}(\lambda x \in D_e . [P(x) \wedge \text{on.the.table}(x) \wedge x = y])]]$$

Example (56) is a function of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$ that can combine with $\llbracket [\text{NP books}] \rrbracket$, which is of type $\langle e, t \rangle$. But what is the motivation for a representation like (56)?

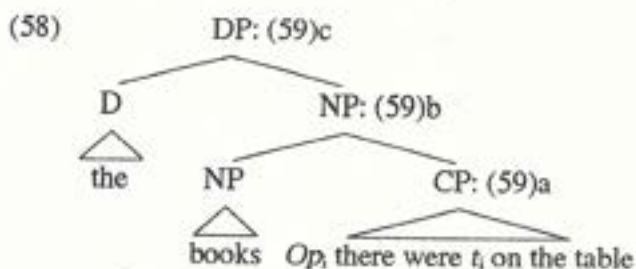
4.3. The Determiner and Negation Restrictions

McNally (1998: 388) claims that "the existential force associated with existential sentences is due to a lexical entailment of the existential predicate". For example, (51)a entails the

introduction of a discourse referent into the domain of the context that corresponds to the instantiation of the property argument $\lambda x[\text{book}(x)]$. I assume that $\lambda x[\text{book}(x)]$ is instantiated by a unique plural individual constituted by the books that there were on the table¹¹. I propose that (56) can be motivated by introducing this lexical entailment into the domain of the *content* of ERs. In particular, I propose that y is a variable over the relevant kinds of individuals in (56). In the spirit of Chierchia (1995: 103-104), I assume that the operation that introduces y in (56) is a static version of *Existential Disclosure* (ED). ED involves adding an equation of the form $x = u$ to a representation like (51)b and abstracting over u :

$$(57) \quad \lambda u[\text{there.be}(\lambda x[\text{book}(x) \wedge \text{on.the.table}(x) \wedge x = u])]$$

Under this assumption, (58) is interpreted as in (59); ED is triggered by a feature $[f]$ of the highest IP that dominates a free variable of type $\langle e, t \rangle$:



- (59) a. $\llbracket [\text{CP } Op_i [\text{IP}[\eta] \text{ there were } t_i \text{ on the table}]] \rrbracket =$
 $\lambda P \lambda u[\text{there.be}(\lambda x[P(x) \wedge \text{on.the.table}(x) \wedge x = u])]$
 b. $\llbracket [\text{NP } \text{books } Op_i [\text{IP}[\eta] \text{ there were } t_i \text{ on the table}]] \rrbracket =$
 $\lambda u[\text{there.be}(\lambda x[\text{book}(x) \wedge \text{on.the.table}(x) \wedge x = u])]$
 c. $\llbracket [\text{DP } \text{the books } Op_i [\text{IP}[\eta] \text{ there were } t_i \text{ on the table}]] \rrbracket =$
 $u[\text{there.be}(\lambda x[\text{book}(x) \wedge \text{on.the.table}(x) \wedge x = u])]$

If u is a variable over either atomic or plural individuals, (59)b is a singleton set because u is unique. If (59)b is a *plural* singleton set, it can be the generator set for a Class I determiner (cf. §3.3)¹². The negation restriction follows from the proposals above because negation is opaque for binding of this discourse referent. [I am not fluent enough in dynamic semantics to explain this restriction formally yet.]

¹¹ In (i), $\lambda x[\text{book}(x)]$ is instantiated by a unique *atomic* individual. Similarly, in (ii) the generalized quantifier meaning of the nominal phrase containing the ER does not have a plural singleton generator set, thus (ii) is ungrammatical:

- (i) There was a book on the table.
 (ii) *the book [(that) there was ___ on the table]

¹² Modulo a type $\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$ version of *the*, i.e., $\lambda Q \lambda P[P(u(Q(x)))]$.

4.4. Reconstruction Effects in ERs

Consider example (60):

- (60) Lenny and Carl are the women [(that) Homer believes __ were on the balcony].

Example (60) is ambiguous because the descriptive content (DC) of *the women* can be interpreted either inside or outside the scope of *believes*. Consider the stories in (61):

(61) a. *Wide Scope Story*

Homer arrives at work and sees Lenny and Carl waving at him from the balcony of the employees' lounge. But unbeknownst to Homer, Lenny and Carl are really women posing as men.

b. *Narrow Scope Story*

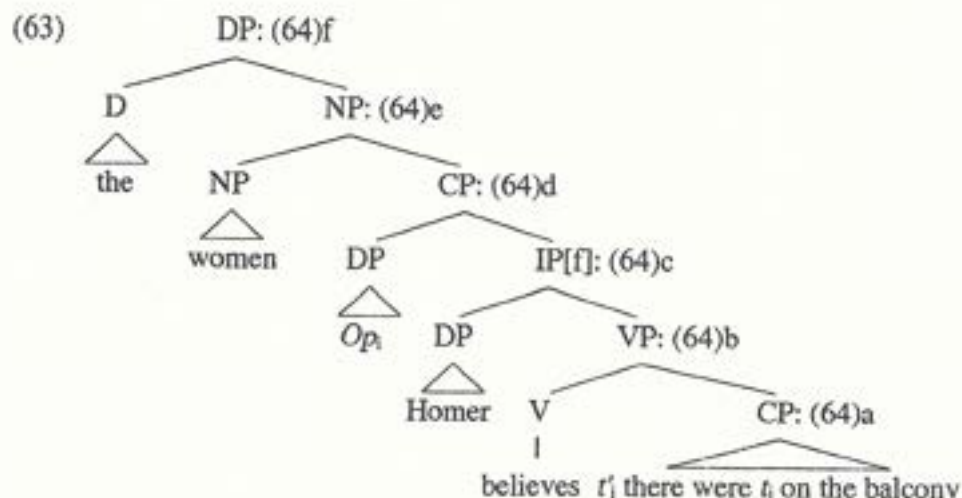
Homer arrives at work and sees two women waving at him from the balcony of the employees' lounge. But Homer is drunk, and the women he sees are Lenny and Carl, who are really men.

Example (60) can describe the situation in (61)a if the DC of *the women* is interpreted outside the scope of *believes*, and can describe the situation in (61)b if the DC of *the women* is interpreted inside the scope of *believes*. But consider example (62):

- (62) Lenny and Carl are the women [(that) Homer believes there were __ on the balcony].

In (62), the DC of *the women* can only be interpreted inside the scope of *believes* (i.e., in the postcopular position of existential *there*), thus (62) can only describe the situation in (61)b. This is a consequence of a representation like (56) that semantically reconstructs the head nominal of an ER to the gap in the postcopular position¹³. Under this assumption, (63) is interpreted as in (64); again, ED is triggered by a feature [f] of the highest IP that dominates a free variable of type $\langle e, t \rangle$:

¹³ See Cresti (1995: 85) and Rullmann (1995: 194-195) for similar analyses of semantic reconstruction in *how many*-questions.



- (64) a. $\llbracket [CP\ t_i\ \text{there were } t_i\ \text{on the balcony}] \rrbracket =$
 $\hat{\text{there.be}}(\lambda x[P(x) \wedge \text{on.the.balcony}(x)])$
- b. $\llbracket [VP\ \text{believes } t_i\ \text{there were } t_i\ \text{on the balcony}] \rrbracket =$
 $\lambda y[\text{believe}(y, \hat{\text{there.be}}(\lambda x[P(x) \wedge \text{on.the.balcony}(x)]))]$
- c. $\llbracket [IP_{[f]}\ \text{Homer believes } t_i\ \text{there were } t_i\ \text{on the balcony}] \rrbracket =$
 $\lambda u[\text{believe}(\text{Homer}, \hat{\text{there.be}}(\lambda x[P(x) \wedge \text{on.the.balcony}(x) \wedge x = u]))]$
- d. $\llbracket [CP\ Op_i\ [IP_{[f]}\ \text{Homer believes } t_i\ \text{there were } t_i\ \text{on the balcony}]] \rrbracket =$
 $\lambda P\lambda u[\text{believe}(\text{Homer}, \hat{\text{there.be}}(\lambda x[P(x) \wedge \text{on.the.balcony}(x) \wedge x = u]))]$
- e. $\llbracket [NP\ \text{women } Op_i\ [IP_{[f]}\ \text{Homer believes } t_i\ \text{there were } t_i\ \text{on the balcony}]] \rrbracket =$
 $\lambda u[\text{believe}(\text{Homer}, \hat{\text{there.be}}(\lambda x[\text{woman}(x) \wedge \text{on.the.balcony}(x) \wedge x = u]))]$
- f. $\llbracket [DP\ \text{the women } Op_i\ [IP_{[f]}\ \text{Homer believes } t_i\ \text{there were } t_i\ \text{on the balcony}]] \rrbracket =$
 $u[\text{believe}(\text{Homer}, \hat{\text{there.be}}(\lambda x[\text{woman}(x) \wedge \text{on.the.balcony}(x) \wedge x = u]))]$

The DC of *the women* semantically reconstructs to the gap in the postcopular position in (64)e. This follows from the proposal that ERs are of type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ and combine semantically with their head nominals through function application.

5. Conclusion

The basic advantage of the proposals above is that they derive the relativizer and determiner restrictions from the semantics of *there*-existential constructions. They also predict the reconstruction effects in §4.4. [The negation restriction is not explained yet.] However, there is an interesting residual phenomenon to discuss. Consider RRs like (65) that involve *whose* and have the same determiner restriction as ERs, but no relativizer restriction (thanks to Sandy Chung for observing the relevance of these facts):

- (65) *the/*most/*some* students [*whose files* there were ___ in the cabinet]

These RRs also have a negation restriction:

- (66) *the students [*whose files* there weren't __ in the cabinet]

But Heim's analysis of the definiteness restriction predicts that (65) is ungrammatical:

[A] question like *Whose drink did you spill?* must be taken to involve a narrow-scope occurrence of *x's drink*. This will count as a strong NP for whatever reason *that woman's drink* counts as strong, a welcome prediction in view of the following judgment.

- (22) *Whose drink is there *t* left on the table? (Heim (1987: 32))

Assuming, following Chomsky (1977, 1995) that examples like (65) involve pied-piping and have logical forms like (67), the problem is that *{x's files}* counts as strong:

- (67) the students [*who* there were *{x's files}* in the cabinet]

However, (65) seems grammatical and seems to be sensitive to the theoretical principle responsible for uniqueness/maximality in ERs¹⁴.

Do RRs like (65) have reconstruction effects with respect to the descriptive content of their heads? Consider example (68):

- (68) Lenny and Carl are the women [*whose undergarments* Homer believes __ were on the balcony].

Example (68) is four ways ambiguous because the DC of *the women* can be interpreted either inside or outside the scope of *believes* and the DC of *whose undergarments* can be interpreted either inside or outside the scope of *believes*. But consider example (69):

- (69) Lenny and Carl are the women [*whose undergarments* Homer believes there were __ on the balcony].

In (69) the DC of *whose undergarments* can only be interpreted inside the scope of *believes* (i.e., in the postcopular position of existential *there*), but the DC of *the women* can be interpreted either inside or outside the scope of *believes*. Thus, examples like (65-69) pattern with ERs with respect to the determiner and negation restrictions, but with RRs with respect to the relativizer restriction and reconstruction effects. I leave the job of working out this separation formally for future research.

¹⁴ Trivially, (65) is not sensitive to the theoretical principle responsible for the type of the relativizer and the gap in ERs.

References

- Barwise, Jon and Robin Cooper: 1981, 'Generalized Quantifiers and Natural Language', *Linguistics and Philosophy* 4: 159-219.
- Beghelli, Filippo and Tim Stowell: 1997, 'Distributivity and Negation', in Anna Szabolcsi (ed.), *Ways of Scope Taking*, Kluwer, Dordrecht: 71-107.
- Carlson, Greg: 1977, 'Amount Relatives', *Language* 53: 520-542.
- Chierchia, Gennaro: 1995, *Dynamics of Meaning*, University of Chicago Press, Chicago.
- Chomsky, Noam: 1977, 'On WH-Movement', in Peter Culicover et al. (eds.), *Formal Syntax*, Academic Press, New York: 71-132.
- Chomsky, Noam: 1995, *The Minimalist Program*, MIT Press, Cambridge, MA.
- Cresti, Diana: 1995, 'Extraction and Reconstruction', *Natural Language Semantics* 3: 79-122.
- Dekker, Paul: 1993, 'Existential Disclosure', *Linguistics and Philosophy* 16: 561-587.
- Grosu, Alexander and Fred Landman: 1998, 'Strange Relatives of the Third Kind', *Natural Language Semantics* 6: 125-170.
- Heim, Irene: 1987, 'Where Does the Definiteness Restriction Apply?', in Eric Reuland and Alice ter Meulen (eds.), *The Representation of (In)Definiteness*, MIT Press, Cambridge, MA: 21-42.
- Ladusaw, William: 1982, 'Semantic Constraints on the English Partitive Construction', *Proceedings of WCCFL* 1: 231-242.
- Link, Godehard: 1983, 'The Logical Analysis of Plurals and Mass Terms', in Rainer Bäuerle et al. (eds.), *Meaning, Use, and Interpretation of Language*, de Gruyter, Berlin: 302-323.
- McNally, Louise: 1992, *A Semantics for the English Existential Construction*, PhD dissertation, University of California, Santa Cruz. Revised version published 1997 by Garland Press, New York.
- McNally, Louise: 1998, 'Existential Sentences Without Existential Quantification', *Linguistics and Philosophy* 21: 353-392.
- Partee, Barbara: 1987, 'Noun Phrase Interpretation and Type-Shifting Principles', in Jeroen Groenendijk et al. (eds.), *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers*, Foris, Dordrecht: 115-143.
- Rullmann, Hotze: 1995, *Maximality in the Semantics of WH-Constructions*, PhD dissertation, University of Massachusetts, Amherst.